# Miter Bend Calculation Report

ASME B31.3-2018 | Internal pressure

## DESIGN EQUATION

ASME B31.3-2018, 304.2.3
q = T - c; r2 = (D - T)/2; theta = angle/(2 joints)
(4a) Pm = S E W q/r2 * q/[q + 0.643 tan(theta) sqrt(r2 q)]
(4b) Pm = S E W q/r2 * (R1-r2)/(R1-0.5 r2)
Multiple: min(4a,4b), theta <= 22.5 deg
Single: (4a) if theta <= 22.5 deg; otherwise (4c)
(4c): use 1.25 instead of 0.643 in (4a)
(5) R1 >= A/tan(theta) + D/2 (empirical A, not allowance c)
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

| | |
|---|---|
| Project | **QA miter** |
| Prepared by | **Device preview** |
| Design reference | **Illustrative coefficients only** |
| Report date | **2026-10-21 16:36** |

## 01 INPUTS

| | |
|---|---|
| Design gauge pressure | **20 bar** |
| Miter cut angle theta | **22.5 deg** |
| Material / product | **A106 Gr.B - user stress** |
| Design metal temperature | **20 C** |
| S at design temperature | **100 MPa; user supplied, A-1/A-1M** |
| E / W | **1 / 1** |
| OD D / segment wall T | **114.3 / 6.02 mm** |
| Total allowances c | **1 mm** |
| Miter weld joint count | **2** |
| Effective radius R1 | **180 mm** |

## 02 RESULTS

| | |
|---|---|
| Governing equation | **Eq. 4a & 4b (multiple miter)** |
| Maximum pressure Pm | **49.4608 bar** |
| Pressure margin Pm-P | **29.4608 bar** |
| Required R1 / wall extension M | **117.5053 / 52.1329 mm** |
| Minimum T meeting pressure | **3.4978 mm; numerical solution** |

**MEETS the implemented internal-pressure equations.**

Provide the required extension M in the actual fabrication geometry; the app does not measure it. Overall design, material suitability, fabrication and other loads require independent verification.

## 04 REFERENCES AND LIMITS

Sources: B31.3-2018 304.2.3, supplied printed pp.22-23; B31.1-2022 104.2.3, supplied printed p.29. Inputs are user supplied. No external-pressure, fatigue, remaining-life or complete code qualification is performed.